$$\mathbf{PX}\psi(x) = -i\hbar\frac{d\big(x\psi(x)\big)}{dx}.$$

To calculate this last expression, we just use the standard rule for differentiating the product $x\psi(x)$. Using this rule, it's easy to see that

$$\mathbf{PX}\psi(x) = -i\hbar x\frac{d\psi(x)}{dx} - i\hbar\psi(x). \qquad (8.28)$$

Now, we'll subtract Eq. 8.28 from Eq. 8.27 to show how the commutator acts on the wave function:

$$[\mathbf{X},\mathbf{P}]\psi(x) = \mathbf{XP}\psi(x) - \mathbf{PX}\psi(x)$$

or

$$[\mathbf{X},\mathbf{P}]\psi(x) = i\hbar\psi(x).$$

In other words, when the commutator [**X**, **P**] acts on *any* wave function $\psi(x)$, all it does is multiply $\psi(x)$ by the number $i\hbar$. We can express this by writing

$$[\mathbf{X},\mathbf{P}] = i\hbar. \qquad (8.29)$$

This in itself is extremely important. The fact that **X** and **P** don't commute is the key to understanding why they are not simultaneously measurable. But things get even more interesting when we compare this equation with Equivalence 8.26, which relates commutators to classical Poisson brackets. In fact, Eq. 8.29 suggests that the corresponding classical Poisson bracket is

$$\{x,p\} = 1,$$